

Homework Gaussian (2026)

- This Homework Assignment consists of six (6) exercises: 4 computational exercises and 2 theoretical questions.
- Please prepare a report containing the answers, solutions and explanations for these exercises.
- Software to use for this report: **Gaussian 16** and **GaussView 6**.
- Deadline to submit the report: **Sunday, 13 September 2026, 23h59m**.
- The report must consist of **a single PDF file** for each student.
- A maximum of 5 (five) upload attempts are allowed before the deadline. The latest uploaded file will be used for verification.
- This Homework Assignment weighs **15%** of the final grade for the CM5235 module.

Instructions:

- Be concise in your answers. Write answers exactly to the point.
- Try to achieve the results individually. Copying someone else's answers, reports, or computational results does not help your own development and understanding.
- Plots must be done by Origin, Excel, Mathematica, Python, or equivalent—no hand-made drawn plots.

Full Name : _____

Student Number: _____

E-mail: _____

Date : _____

Signature : _____

Exercise 1**(5 marks)****Study of the ground state HCl diatomic molecule with different computational methods**

Evaluate the spin restricted Hartree-Fock ground state energy for HCl molecule, interatomic distance $R=1.275 \text{ \AA}$. A typical example of input is given below.

```
%chk=C:/HCl.chk <please update path accordingly>
%mem=6MW
%nproc=1
#p hf/sto-3g guess=save geom=connectivity scf=tight

Title=Calculation of the ground state energy with STO-3G basis set

0 1
Cl
H          1          B1

B1          1.275000000

1 2 3.0
2
```

In the above input, you need to update “%chk=C:/HCl.chk” line so that it points to your path in \$HOME on HPC.

Run this calculation and find in the obtained log file the following information:

- SCF energy
- Electric dipole moment
- Total number of molecular orbitals (occupied and virtual /empty) and their energies
- Mulliken charges

1) **Optimize the molecular structure** and **evaluate vibrational frequency** at the using the following basis sets: sto-3g, cc-pVDZ, cc-pVTZ, cc-pVQZ, 6-31g and 6-311+g(d). Make a table comparing the obtained results: optimal bond length, vibrational frequency, number of orbitals, number of basis functions, total energy, HOMO-LUMO gap.

Compare the obtained calculated results (bond length, vibrational frequency, Entropy, Enthalpy, momenta of inertia, etc.) for the ground state with the available experimental results, which can be retrieved from <https://cccbdb.nist.gov/exp2x.asp> website (search for “HCl” in the search field).

2) Draw the orbital energy diagram and plot occupied molecular orbitals for HCl molecule for the RHF/6-311+g(d) method.

Exercise 2**(5 marks)****Study of the Potential Energy Surface for H-F diatomic molecule**

Calculate the *potential energy surface* for **H-F** diatomic molecule using the following methods:

- spin-unrestricted HF/ cc-pVTZ basis set
- spin-unrestricted B3LYP/ cc-pVTZ basis set
- spin-unrestricted CCSD / cc-pVTZ basis set
- spin-unrestricted CCSD(T) / cc-pVTZ basis set

Scan the interatomic distance from $0.50 \text{ \AA}$ till $10.00 \text{ \AA}$, step size of $0.01 \text{ \AA}$.

- Report the results in the form of a plot $E=f(R)$, where R is the inter-atomic distance. Make a single plot combining the four potential energy surfaces.
- What is the most optimal bond length, according to this computational method?
- What is the energy required to break the **HF** bond, for each method?
- How do your calculated values correlate with the experimental bond dissociation energy of 568.6 kJ/mol? Which computational method is more accurate?

Exercise 3

(5 marks)

Bond dissociation energy

Draw the **F-CH₂-Cl** molecule in GaussView and perform a **structural optimization** using the following methods:

- using HF/STO-3G method (minimal accuracy)
- using B3LYP/cc-pVDZ method (reasonable accuracy)
- using CCSD/cc-pVTZ method (high accuracy)

- Compare the **total energies** of the ground state for all these methods
- for the B3LYP/cc-pVDZ method, **plot the electronic density together with electrostatic potential** for this molecule
- Calculate the **bond dissociation energy** for C-H and C-F bonds in the above molecule using the B3LYP/cc-pVDZ computational method. What computational strategy do you use for this purpose? Which of the two atoms is the easiest to be removed?

Exercise 4

(5 marks)

Theoretical question #1

Describe the basic approximations used in the Hartree-Fock Self Consistent Field Method:

- Born Oppenheimer Approximation
- Orbital Approximation
- Mean Field Approximation

How do these approximations simplify the computational problem?

Exercise 5

(5 marks)

Theoretical question #2

Describe the covalent chemical bonding on the example of H₂ molecule. Write down the Slater Determinant for the ground state and give an estimate of the total energy using the method explained in Lecture 3 (with h , J , K). Write down the energies of all excited Slater Determinants that can be formed using the minimal basis set on each of the H atoms. Explain the Coulomb and Exchange integrals.

Exercise 6

(5 marks)

Excited states and spectra

Draw the **CFH₂-OH** molecule in GaussView and perform a **structural optimization** using the B3LYP/cc-pVTZ method. Further, calculate the first 5 (five) excited states using the following methods:

- Time-dependent DFT calculations that employ the Tamm-Dancoff approximation (TDA keyword, <https://gaussian.com/td/>)
- CIS method (<https://gaussian.com/cis/>, https://people.chem.ucsb.edu/kahn/kalju/chem126/public/elspect_cis.html)

For both these calculations, keep the same basis set.

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Make a table to compare the two sets of excited energies.

Plot the UV/VIS spectra for this molecule using the obtained results (e.g. see here <https://gaussian.com/uvvisplot/> and <https://www.youtube.com/watch?v=XjbbRI8l4AQ>).